

Reorganization of Cortico-Cerebellar Networks Reflects Progression from Familial Risk to Psychotic Illness

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INTRODUCTION

- Altered effective connectivity (EC) within brain networks is a core feature of psychosis, particularly within the default mode (DMN), central executive (CEN), and salience (SN) networks¹
- While first-degree relatives of individuals with psychosis share genetic risk, they do not express the full clinical phenotype, allowing differentiation between risk markers and illness expression²
- Here, we examined EC across cortical and cerebellar representations of the triple network in healthy controls, individuals with schizophrenia and bipolar disorder with psychosis, and their relatives. We hypothesized that:
 - Relative** would show intermediate connectivity patterns consistent with familial vulnerability
 - Individuals with psychosis** would exhibit reorganized cortico-cerebellar connectivity, reflecting transition to clinical illness

AIMS

- Using effective connectivity, determine if the cerebellar contributions to the triple network (salience, default mode, and control) is altered in subjects with **schizophrenia (SZ)** or **bipolar disorder with psychosis (BPp)** compared to **healthy controls (HC)**
- Establish whether patterns of connectivity changes can be predicted by those with familial risk

ACQUISITION AND PROCESSING

- Resting-state fMRI data for individuals with **schizophrenia (n = 68)**, **bipolar disorder with psychosis (n = 32)**, their first-degree relatives (**SZrel, n=43**; **BPprel, n=23**) and **healthy controls (n = 42)** obtained from the Psychosis Human Connectome Project³
- Siemens 3T Prism scanner (TR = 800ms, voxel size = 2 × 2 × 2mm)
- ENIGMA HALPipe used for fMRI preprocessing and quality assessment
- Extracted BOLD signal time series for 400 regions using the **Schaeffer combined atlas**, including **FreeSurfer** and **Buckner 2011 17-network** atlases⁴
 - Three cerebellar subregions, bilateral putamen and caudate, bilateral anterior insula and cingulate, and bilateral dorsolateral, ventrolateral, and medial prefrontal cortex (PFC) were included

METHODS

Group Iterative Multiple Model Estimation

GIMME utilizes key features of unified structural equation modeling (uSEM)⁵

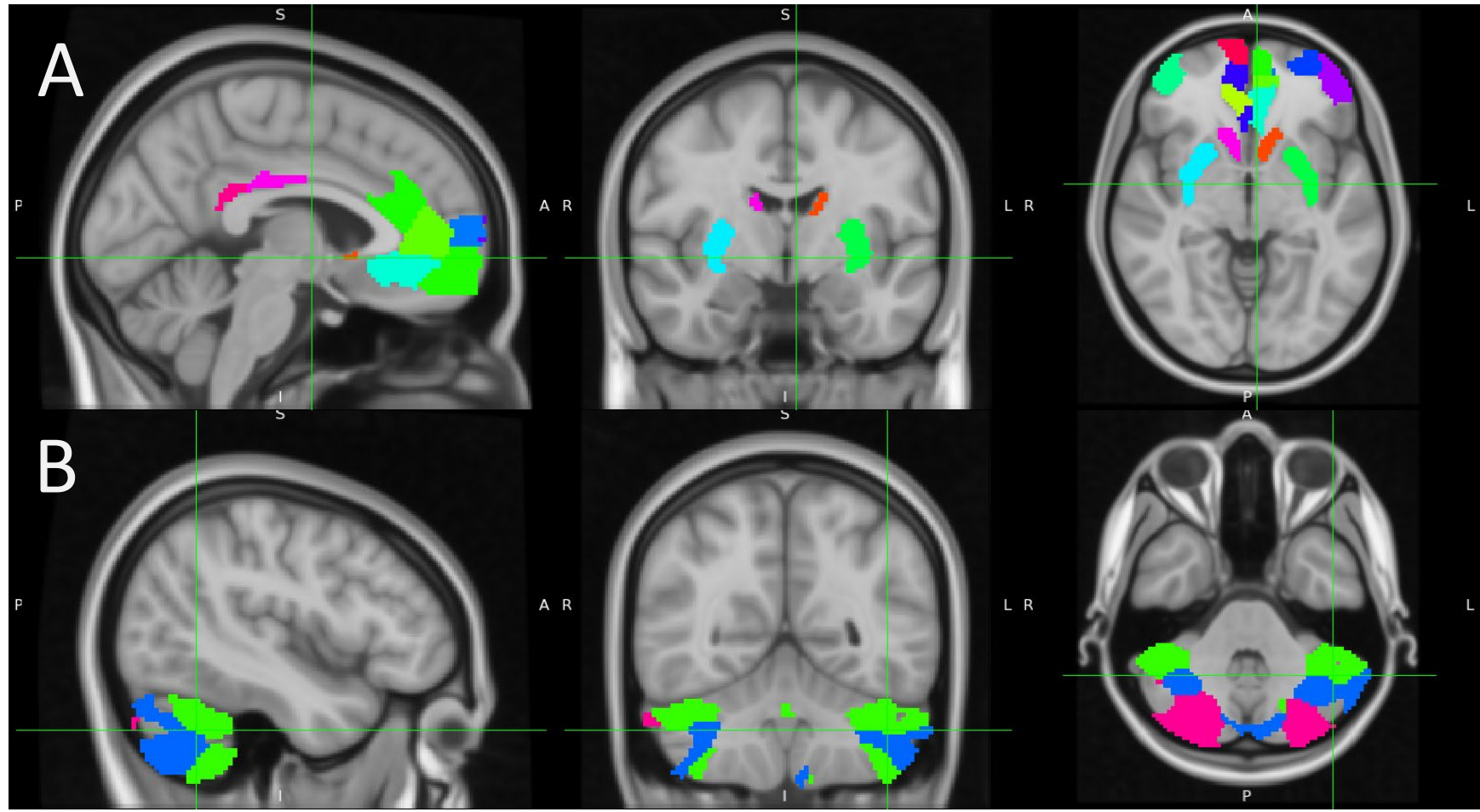
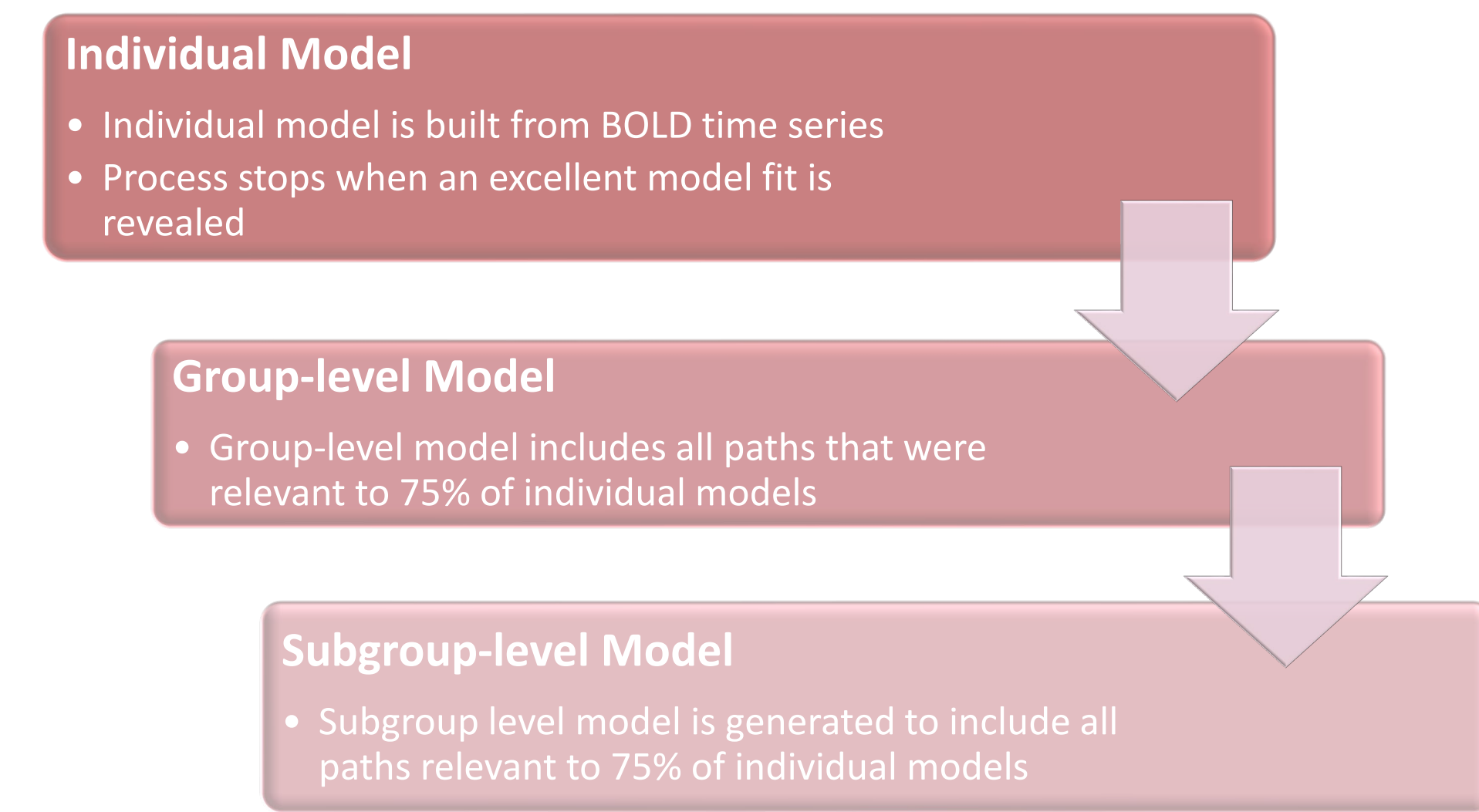
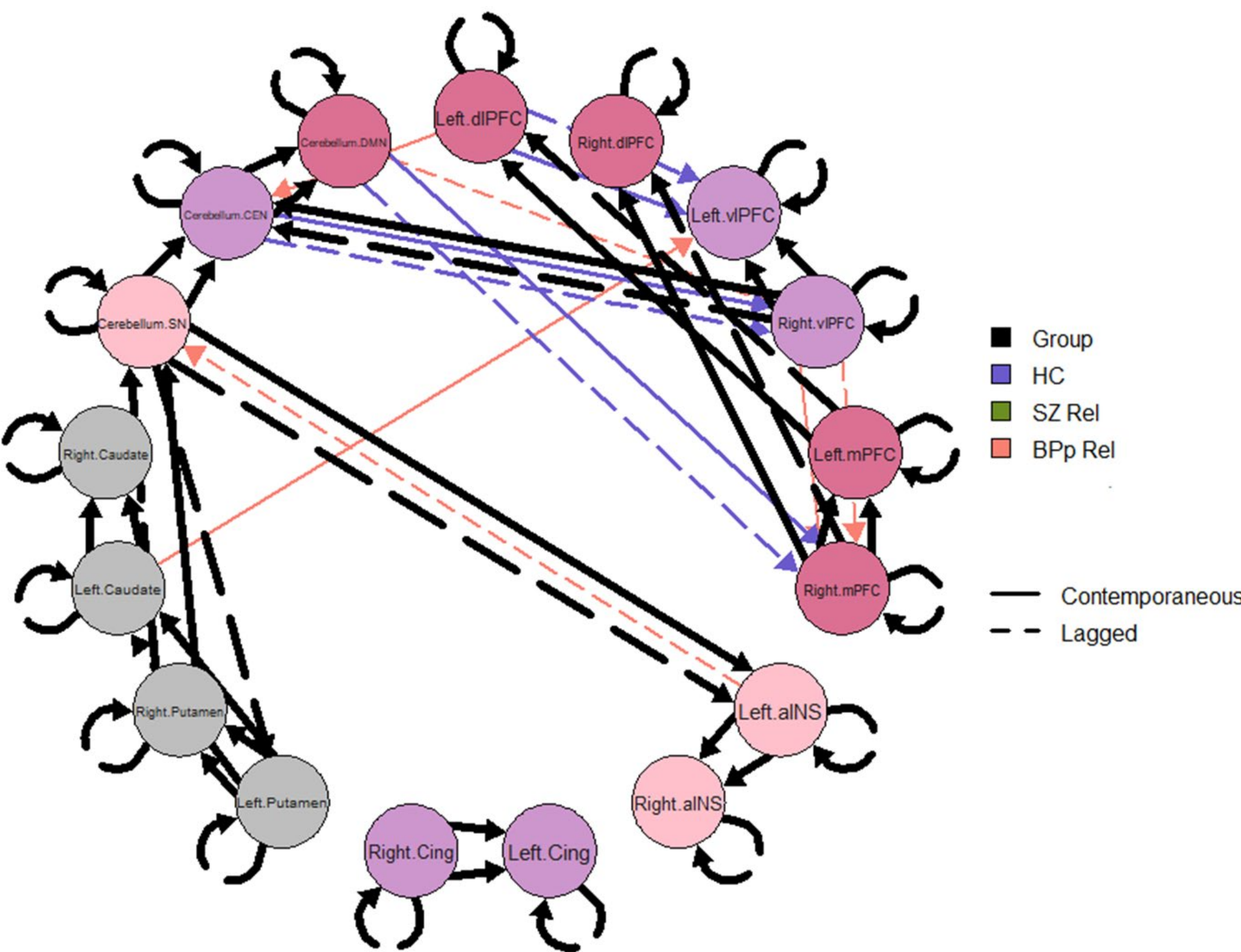


Figure 1: A) Bilateral dorsolateral, ventrolateral, and medial prefrontal cortex, bilateral anterior insula, and bilateral cingulate, were independently averaged from 4-6 ROIs from the Schaeffer atlas. Bilateral putamen and caudate were identified using the Freesurfer atlas. B) The Buckner 2011 atlas was used to select cerebellar representations of the DMN, SN, and CEN. Cerebellar networks correspond to Crus I and Lobule VIIb.

RESULTS

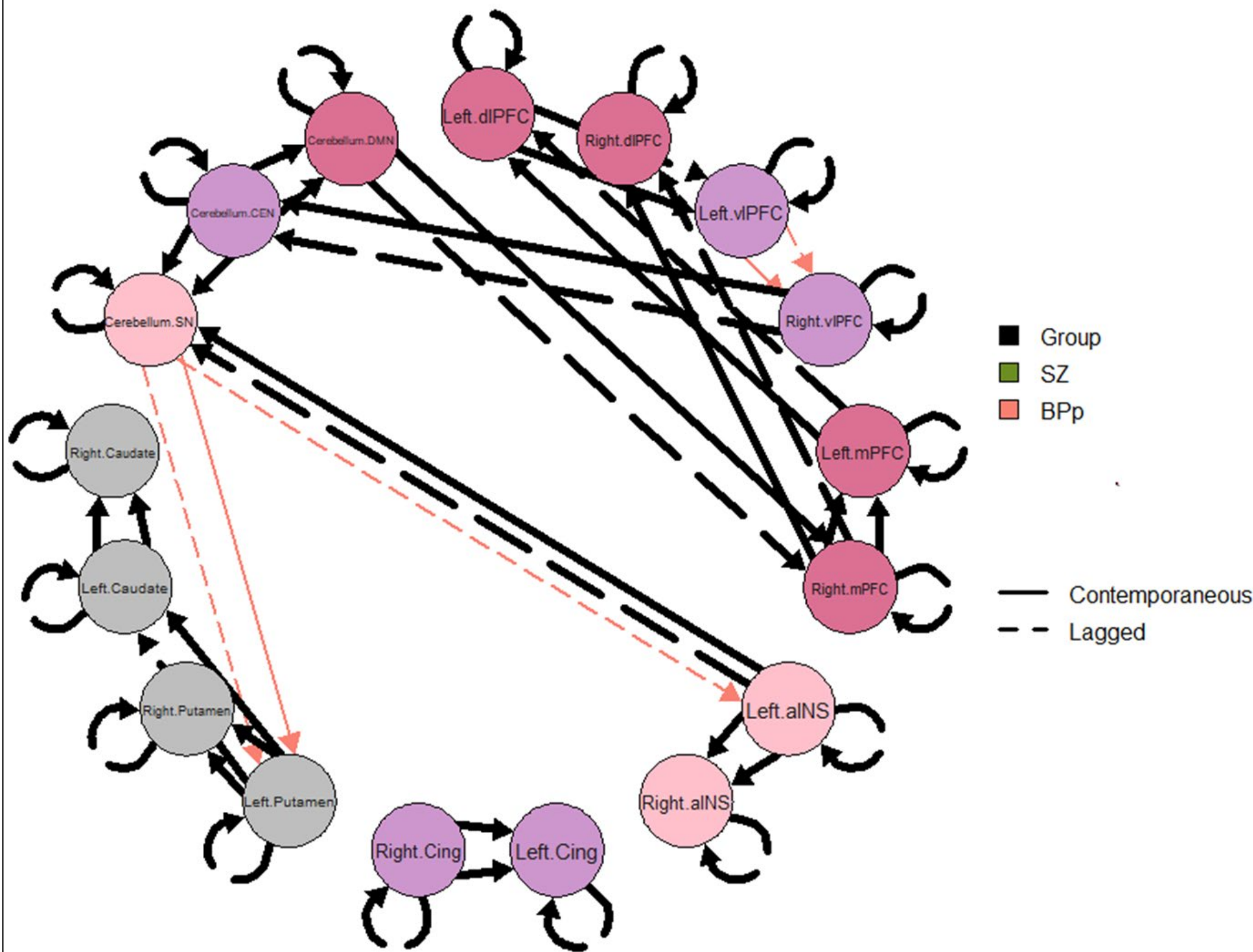
Familial Risk is Associated with Altered Cortico-Cerebellar Connectivity and Reduced Prefrontal Integration



Path_ID	interpretation	estimate	p_fdr
Right.vIPFC~Cerebellum.CEN	HC < BPprel; HC < SZrel	-0.335; -0.376	>0.001; >0.001
Right.vIPFC~Cerebellum.CENlag	HC > BPprel; HC > SZrel	0.341; 0.377	>0.001; >0.001
Left.aINSlag	BPprel > SZrel	0.014	0.04

Figure 2: Supervised clustering identified distinct connectivity patterns across healthy controls (HC) and relatives (SZrel, BPprel). Compared to HCs, relatives showed altered prefrontal integration and cerebellar CEN influence on the vIPFC. Specifically, relatives showed higher lagged connectivity from the right vIPFC to cerebellum CEN, but reduced contemporaneous connectivity of the same path, compared to healthy controls. BPprel showed greater autoregressive connectivity of the left anterior insula than SZrel.

Diagnosis-Specific Differences in Cortico-Cerebellar Connectivity Reflects Divergent Temporal Dynamics in Psychosis



Path_ID	interpretation	estimate	p_fdr
Right.vIPFC~Cerebellum_CEN	BPp > SZ	0.104	0.046
Cerebellum_CEN~Cerebellum_DMN	BPp > SZ	0.085	0.035
Cerebellum_CEN~Cerebellum_DMNlag	BPp < SZ	-0.082	0.035
Left.aINS~Cerebellum_SN	BPp > SZ	0.117	0.008
Left.aINS~Cerebellum_SNlag	BPp < SZ	-0.131	0.004

Figure 3: SZ and BPp exhibited distinct connectivity profiles. BPp uniquely showed enhanced prefrontal connectivity and cerebellar SN projections to the striatum and insula. Compared to SZ, BPp showed stronger contemporaneous connectivity between cerebellar and cortical regions. BPp also exhibits stronger coupling within cerebellar networks. In contrast, SZ shows stronger lagged connectivity, both within the cerebellum and between the cerebellum and insula.

CONCLUSIONS AND FUTURE DIRECTIONS

- Compared to HCs, relatives showed
 - reduced prefrontal connectivity
 - reversed cerebellum.CEN influences on the right vIPFC
 - enhanced striatal-insular coupling
- SZ shows stronger lagged connectivity than BPp, indicating greater temporal delays in network dynamics
- Distinct diagnosis-specific pathways indicate heterogenous network reorganization across psychosis subtypes
- The Cerebellum.SN and anterior insula formed a robust, bidirectional circuit that consistently shows altered connectivity in psychosis, suggesting disrupted integration of salience and predictive processing
 - This circuit represents a key locus linking familial vulnerability to clinical expression

Within-Family Paired Comparison Reveals Cerebellum-Anterior Insula Loop is a Core Circuit in Psychosis

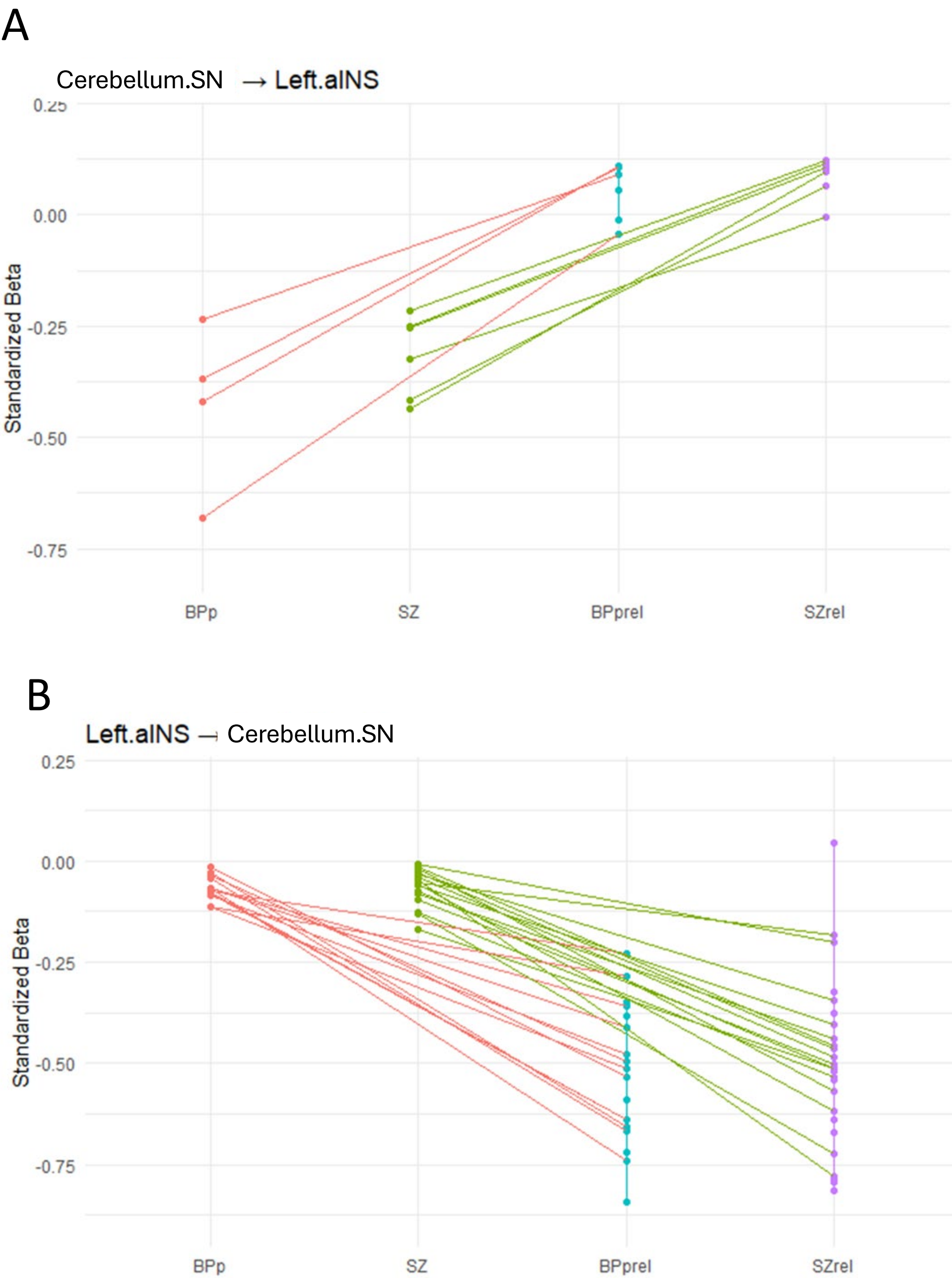


Figure 4: Standardized beta coefficients for bidirectional Cerebellum.SN-Left anterior insula connections across groups. Values reflect within-family paired comparisons between individuals with psychosis (SZ, BPp) and their matched first-degree relatives (SZrel, BPprel). SZ and BPp exhibit A) negative connectivity from Cerebellum.SN → Left.aINS (FDR-corrected $p > 0.001$), whereas relatives exhibit near-zero to positive influence, and B) increased connectivity from Left.aINS → Cerebellum.SN (FDR-corrected $p > 0.001$) compared to their matched relatives

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